Fundamentals 20

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1. Since $f(x_n)$ is Cauchy, it converges to some $y \in \mathbb{R}$ by the completeness of $\mathbb{R}$. So $\lim_{n\to\infty} f(x_n) = y$, and since limits are unique when they exist, y is unique. So defining F as $F(x) = y$ when $x \notin \mathbb{Q}$ guarantees uniqueness.

2. Since f is uniformly convergent, for all $\varepsilon > 0$ there exists $\delta > 0$ such that, for all x_n, x_m, if $|x_n - x_m| < \delta$ then $|f(x_n) - f(x_m)| < \varepsilon$. Also, since x_n is Cauchy, given any δ, there exists N such that if $m, n > N$ then $|x_n - x_m| < \delta$. So if $m, n > N$ then $|x_n - x_m| < \delta$, so $|f(x_n) - f(x_m)| < \varepsilon$.

3. Let (x_n) be any Cauchy sequence with terms in $(0, 1)$ converging to 0. Since f is uniformly continuous, $(f(x_n))$ is Cauchy, so $\lim_{n\to\infty} x_n = \lim_{x_n\to 0} f(x_n)$ exists. Since x_n is arbitrary, $\lim_{x\to 0} f(x)$ exists. So we can define $f(0) = \lim_{x\to 0} f(x)$. By a similar argument, we can define $f(1) = \lim_{x\to 1} f(x)$.

4a. Let $x \in \mathbb{R}$ and $\varepsilon > 0$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there's $y \in \mathbb{Q} \cap \left(\frac{x}{\pi+1} - \frac{\varepsilon}{\pi+1}, \frac{x}{\pi+1} + \frac{\varepsilon}{\pi+1}\right)$. So $\pi y \in \mathbb{R}\backslash\mathbb{Q} \cap (x - \varepsilon, x + \varepsilon)$.

4b. Let $x \in \mathbb{R}$ and $\varepsilon > 0$. Let $k > \log_2 \frac{1}{\varepsilon} - 1$, so that $2^{k+1}\varepsilon > 1$. We have that

$$2^k(x + \varepsilon) - 2^k(x - \varepsilon) = 2^{k+1}\varepsilon > 1.$$

Therefore, there is an integer n between $2^k(x - \varepsilon)$ and $2^k(x + \varepsilon)$, so

$$2^k(x - \varepsilon) < n < 2^k(x + \varepsilon),$$

$$x - \varepsilon < \frac{n}{2^k} < x + \varepsilon,$$

as desired.

4d. Let $x \in \mathbb{R}$ and $\varepsilon > 0$. Let $m > e^{1/\varepsilon}$ be an integer, so that $\frac{1}{\log m} < \varepsilon$. Then let n be the least integer such that $\frac{\log n}{\log m} \geq x$, so that $\frac{\log(n/e)}{\log m} < x$. Then

$$x \leq \frac{\log n}{\log m} = \frac{\log(n/e) + \log e}{\log m} = \frac{\log(n/e)}{\log m} + \frac{1}{\log m} < x + \varepsilon.$$

So if we let $y = \frac{\log n}{\log m}$, then $y \in (x - \varepsilon, x + \varepsilon)$, as desired.

5. Let $f_n(x) = x^n$. Then $\lim_{n\to\infty} f_n(x) = 0$ if $x \in [0, 1)$, but $\lim_{n\to\infty} f(1) = 1$. So f is not continuous at 1.

6.

$$-\sin\left(\sqrt{1+\sin(e^x)}\right)\frac{1}{2\sqrt{1+\sin(e^x)}}\cos(e^x)e^x.$$

7.

$$\left(\frac{f}{g}\right)'(x) = \left(f(x)\left(\frac{1}{g(x)}\right)\right)' = f'(x)\frac{1}{g(x)} + f(x)\left(\frac{1}{g(x)}\right)'$$

$$= \frac{f'(x)}{g(x)} + f(x)\left(-\frac{1}{(g(x))^2}\right)g'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

8. We have that

$$(s_1 \cdots s_n)^{1/n} = \left(\prod_{k=1}^{n} s_k\right)^{1/n} = \exp\left(\frac{1}{n}\sum_{k=1}^{n}\log(s_k)\right).$$

By Proposition 20.10,

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\log(s_k) = \lim_{n\to\infty}\log s_n = \log L.$$

So

$$\lim_{n\to\infty}\exp\left(\frac{1}{n}\sum_{k=1}^{n}\log(s_k)\right) = \exp\left(\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\log(s_k)\right) = \exp(\log L) = L.$$

So the geometric mean converges to L.

If we let $s_k = 0$ for any k then the geometric mean converges while the arithmetic mean may not.